

Spatiotemporal coherence of MHT in the Atlantic Ocean

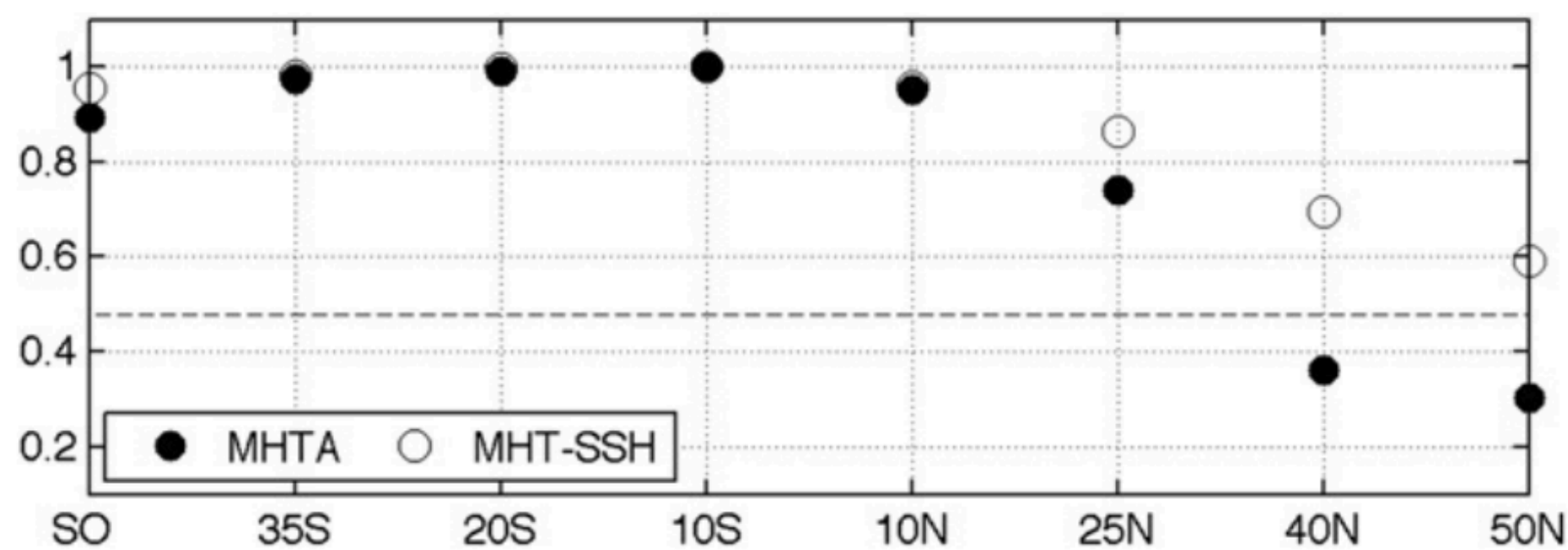
1. Event : breakdown/split of coherence around 2009

By Lea Sophie Dietz

Literature says:

Kelly et al. (2014):

- Coherence south of 40°N up to SO
- Timeperiod: 1993-2011

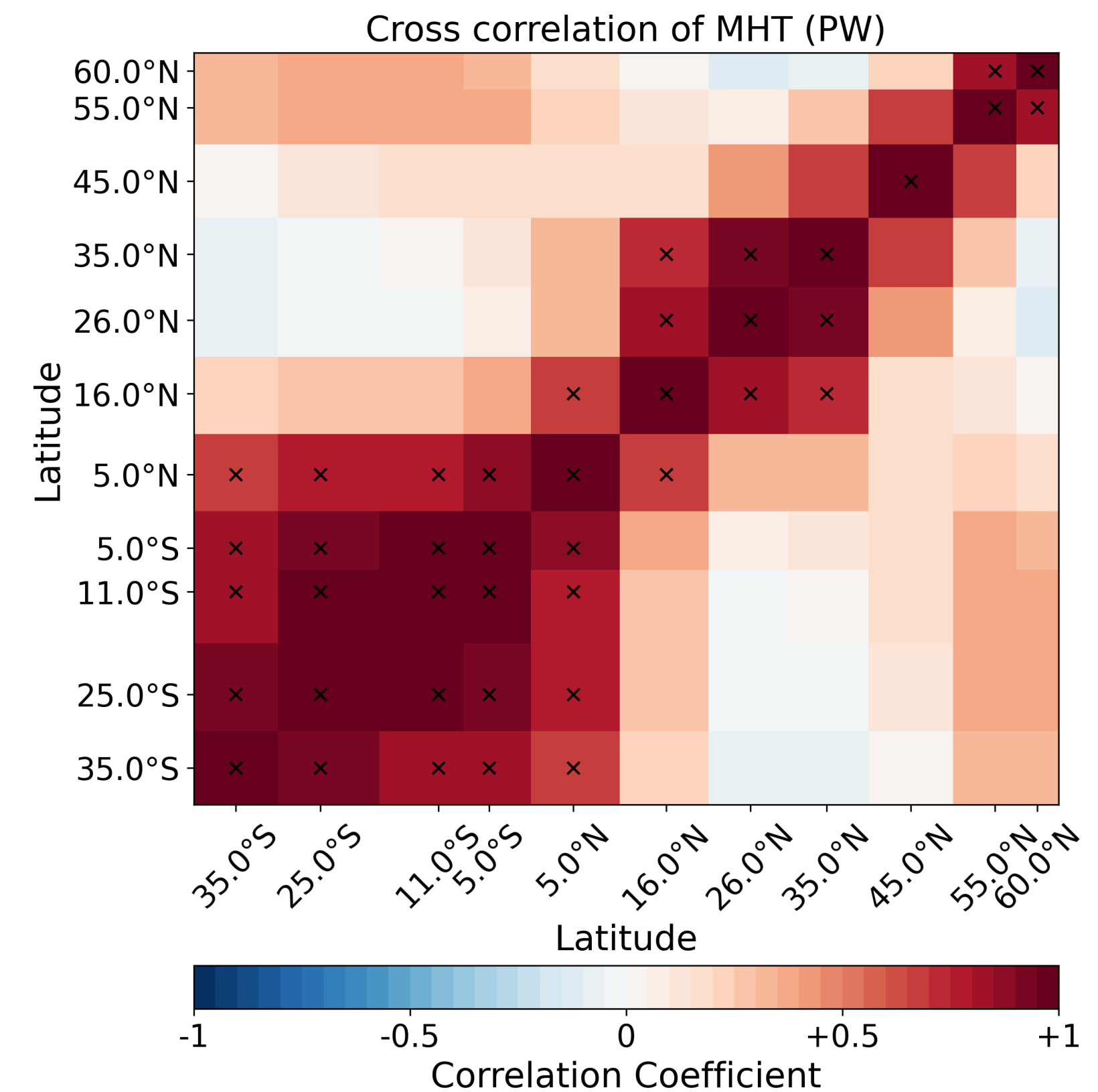


When is this split from coherence up to 40°N and 2 covarying blocks happening and what is causing it?

Calafat et al. (2025):

- 2 Covarying blocks:
- 35-16°N and 5°N-35°S
- Timeperiod: 2004-2020

Cross correlation of MHT anomalies from Calafat dataset:



Idea:

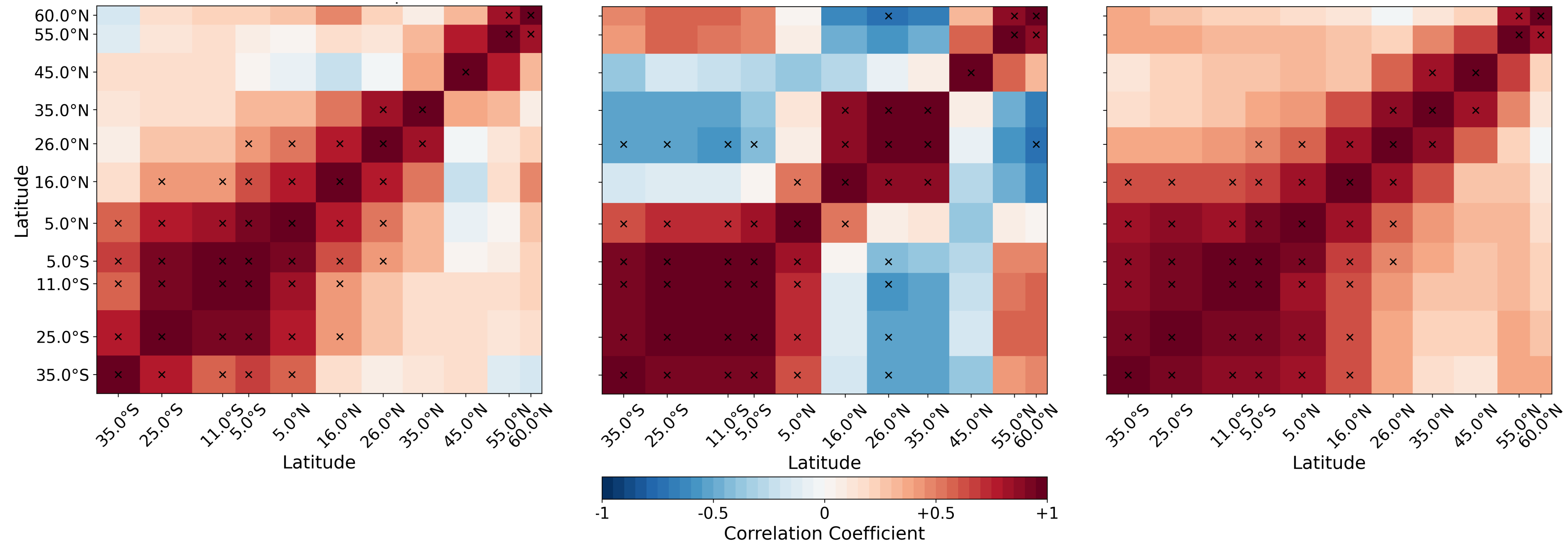
- **split timeseries into before and after split year**
 - **plot gif sliding the split year through the timeseries**
- This led me to distinguish 3 time periods**

Split whole timeseries into 3 periods:

Pre 2009

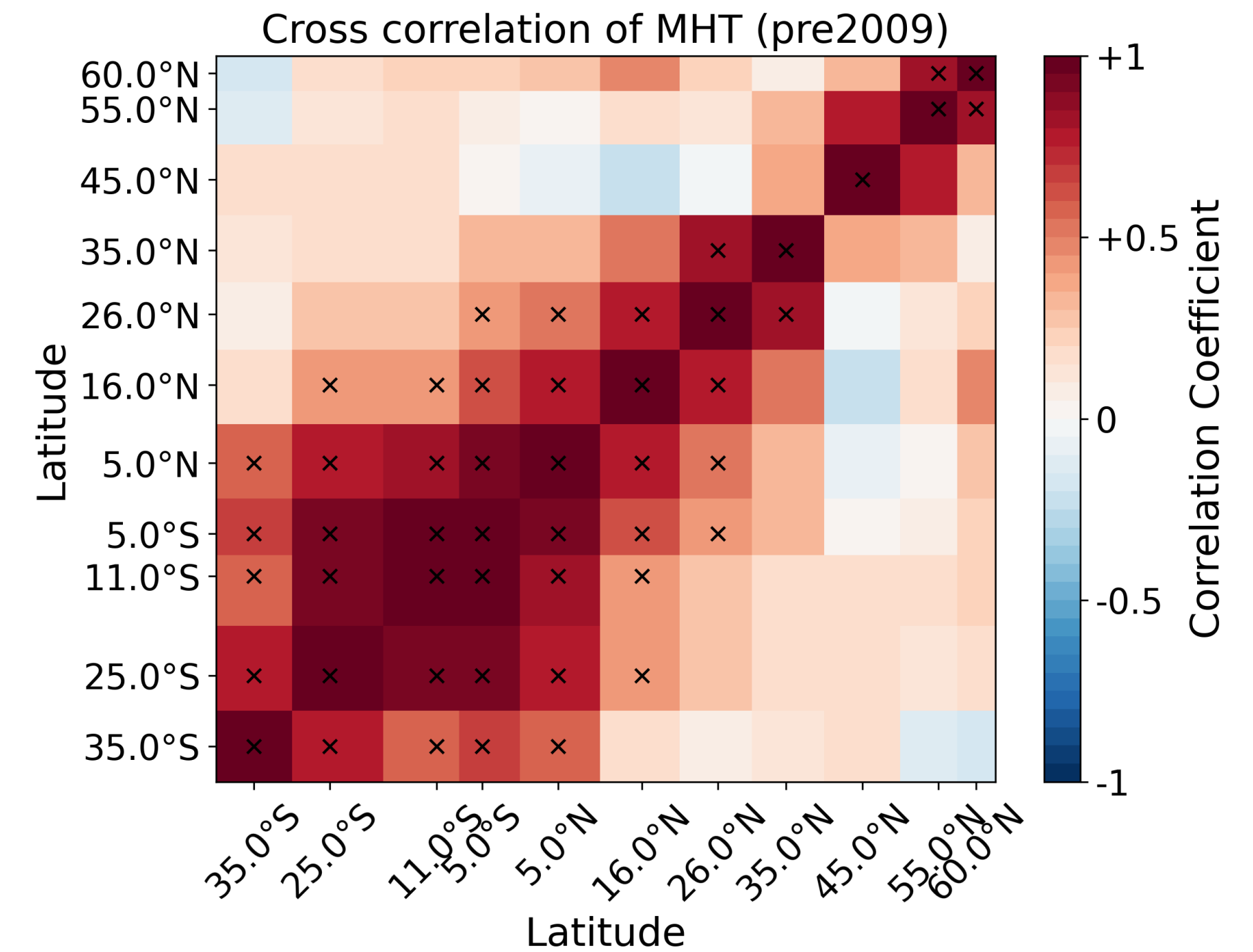
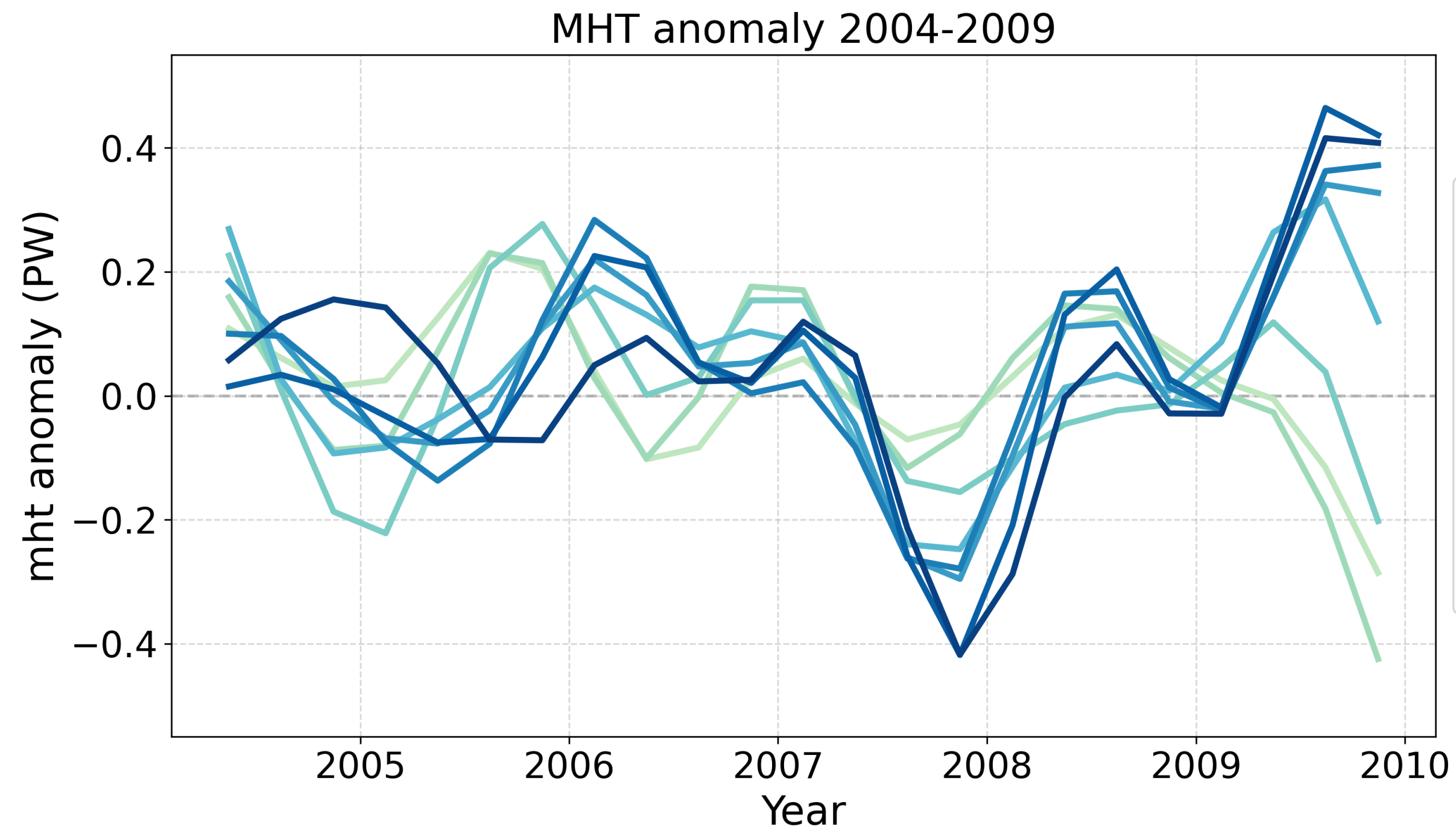
2009-2013

Post 2013



- Before 2009: high correlation of MHT anomalies between latitudes throughout Atlantic south of 35°N
- Then pattern shifts and moves southwards, splits into two boxes around 2009
- Two covarying boxes stays until 2013

MHT Coherence before 2009

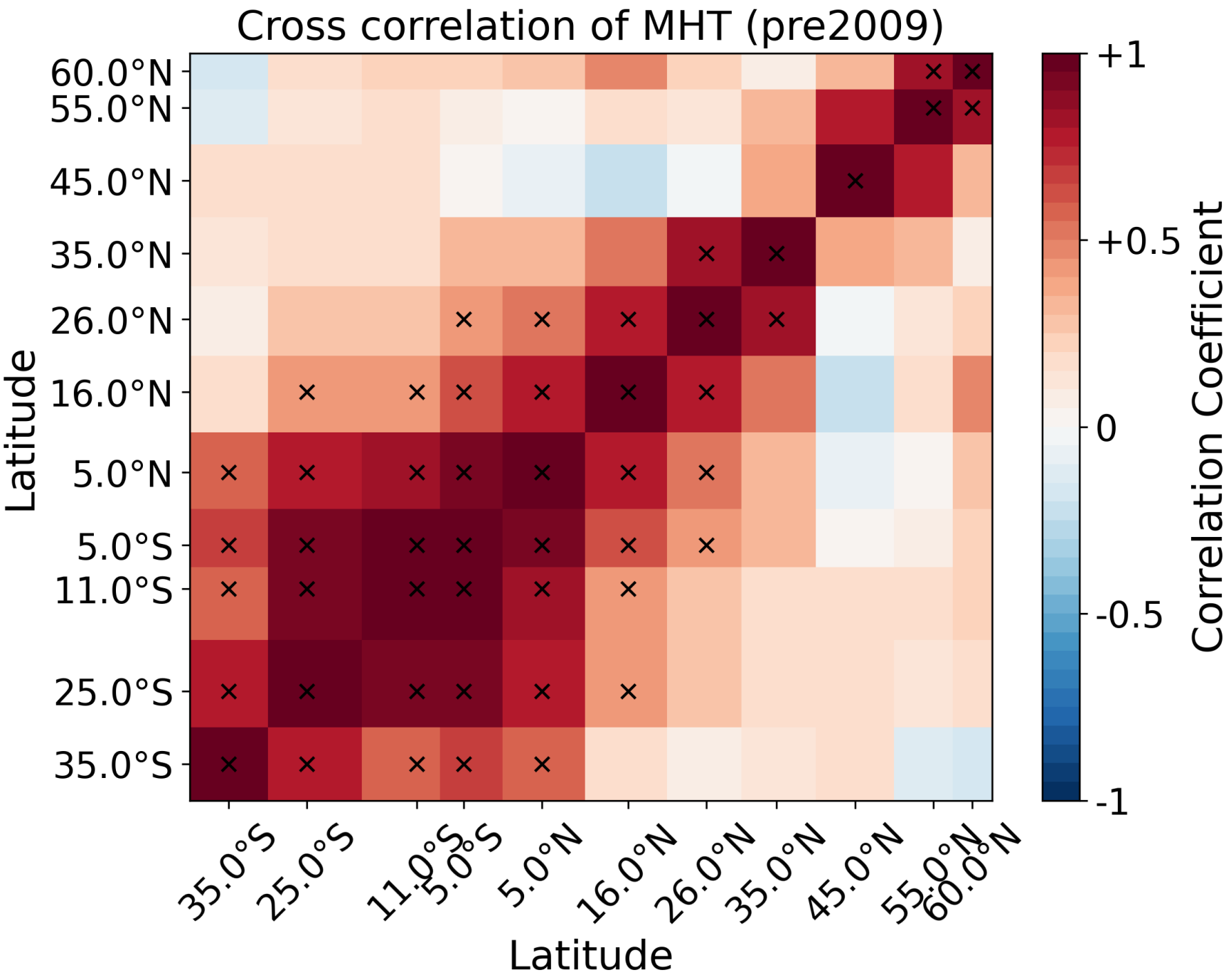
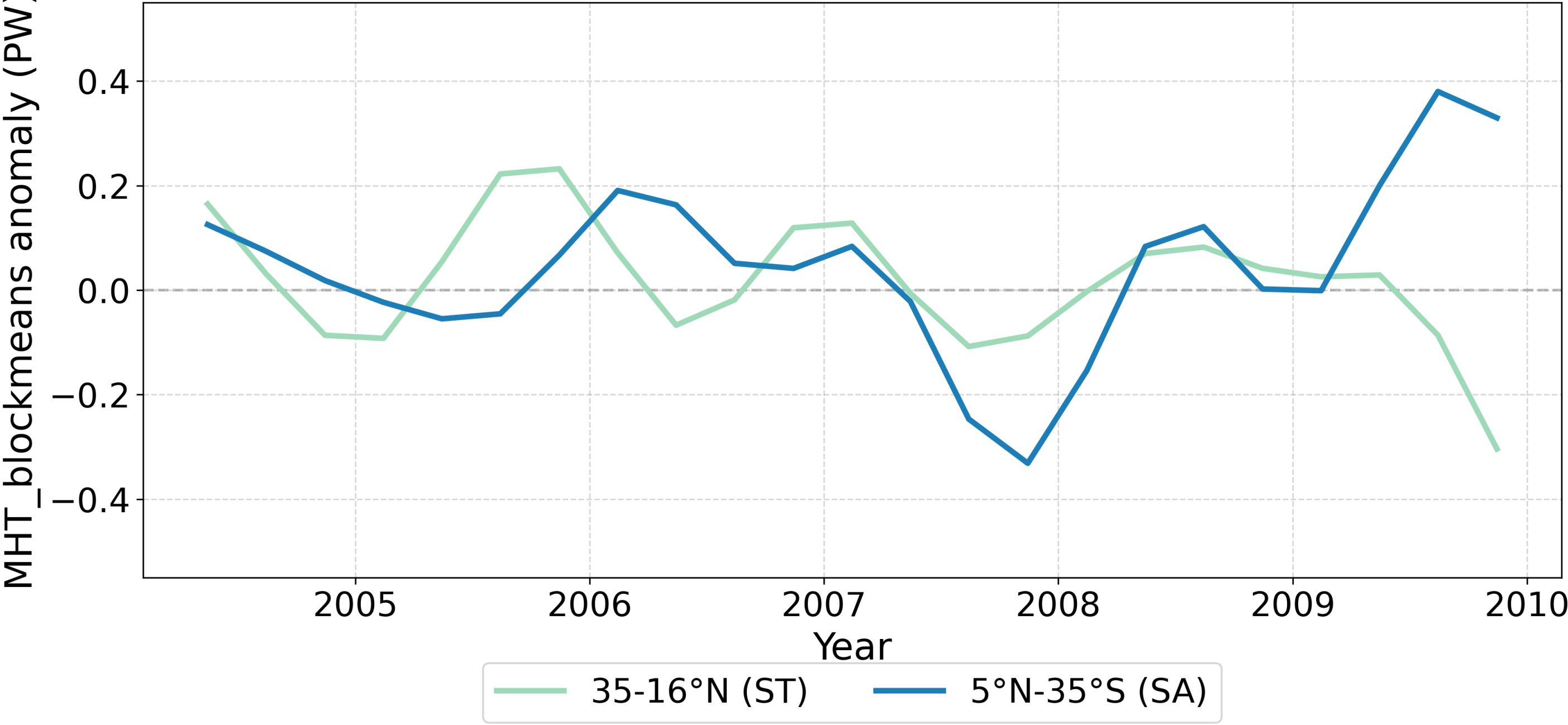


Focus on latitudes south of 45°N: **35°N - 35°S**

- Signal mostly **in phase** (especially from 2007)

MHT Coherence before 2009

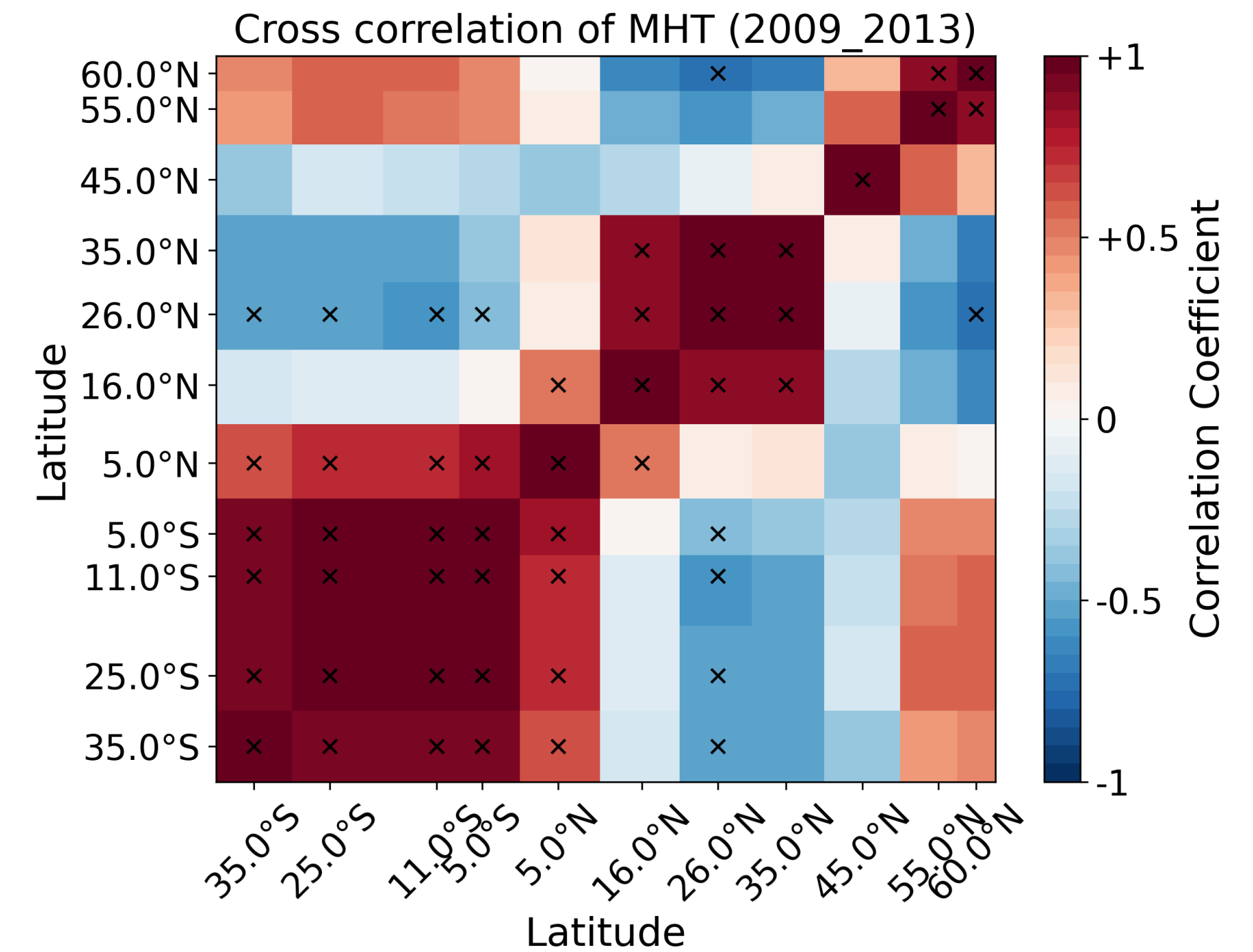
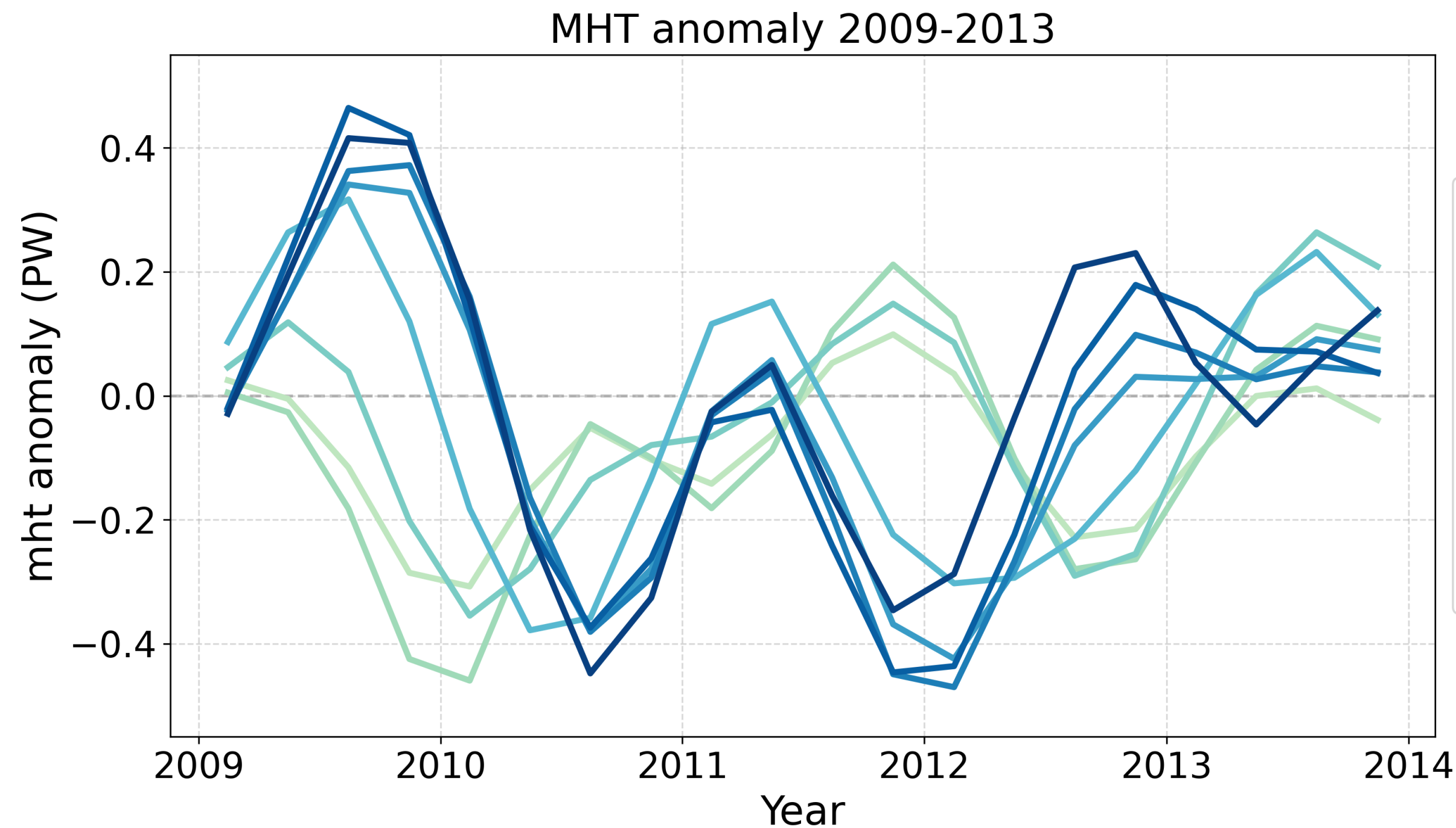
MHT block-mean anomaly 2004-2009



Focus on latitudes south of 45°N: **35°N - 35°S**

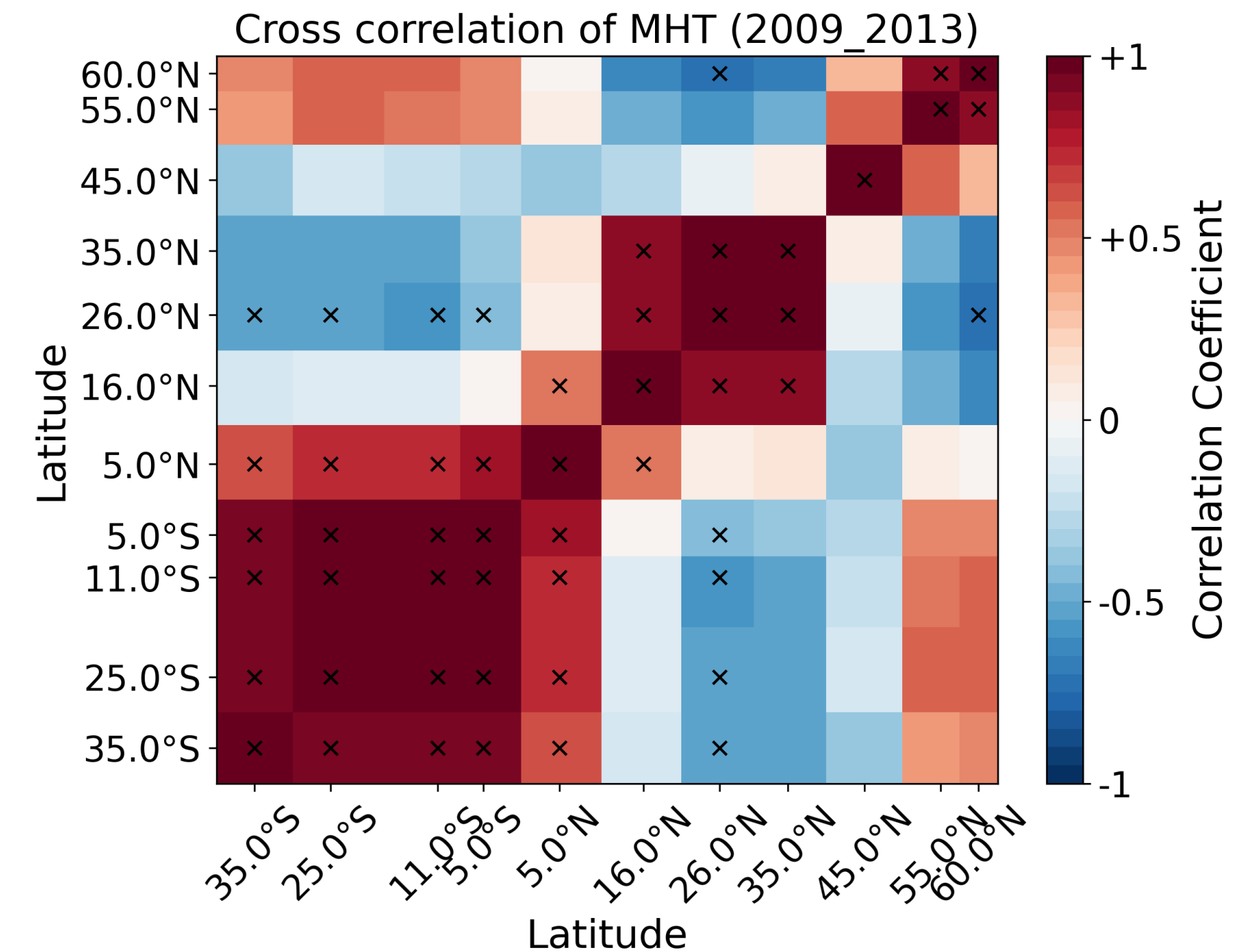
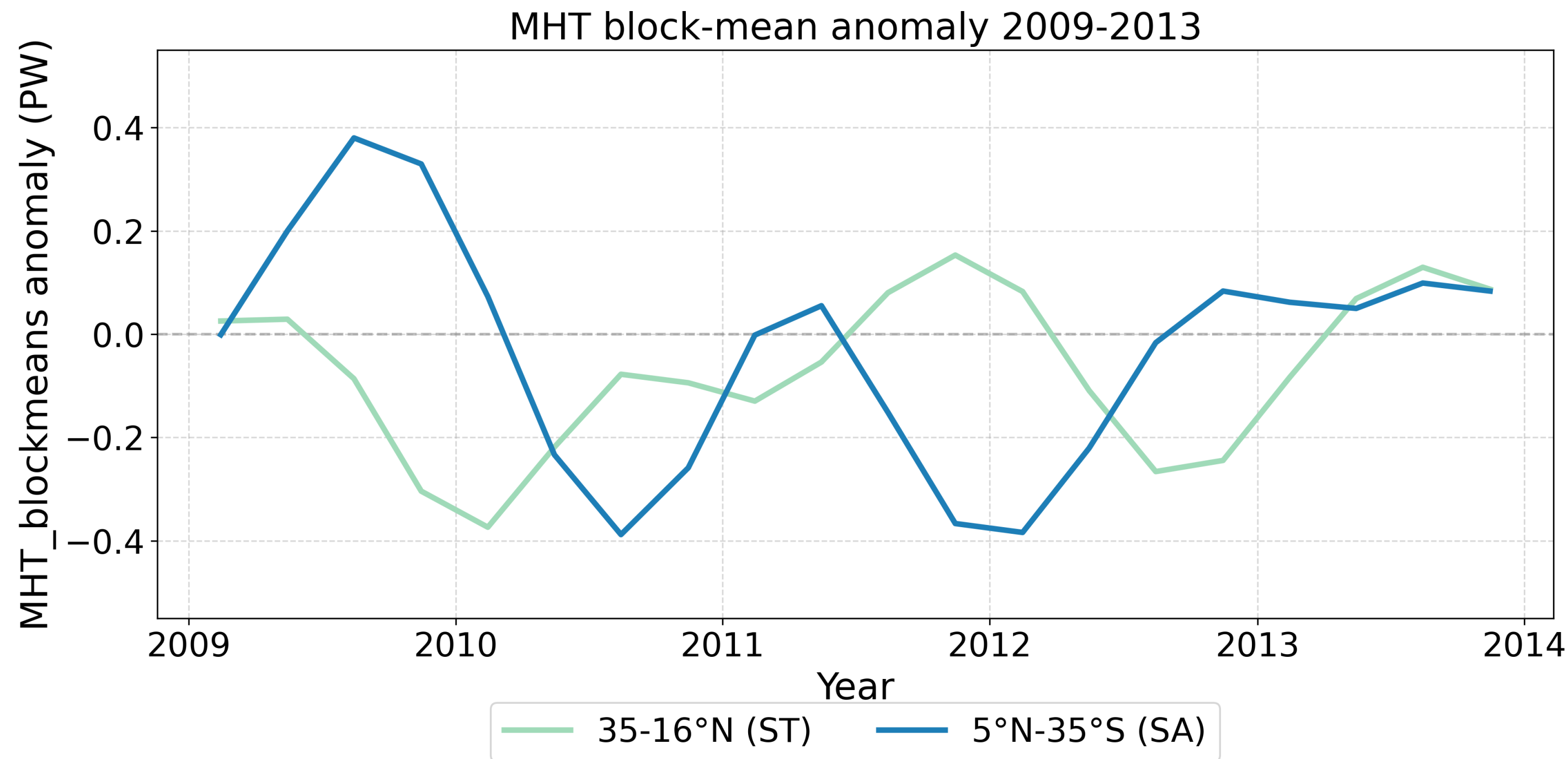
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MHT Coherence / correlation break 2009 - 2013



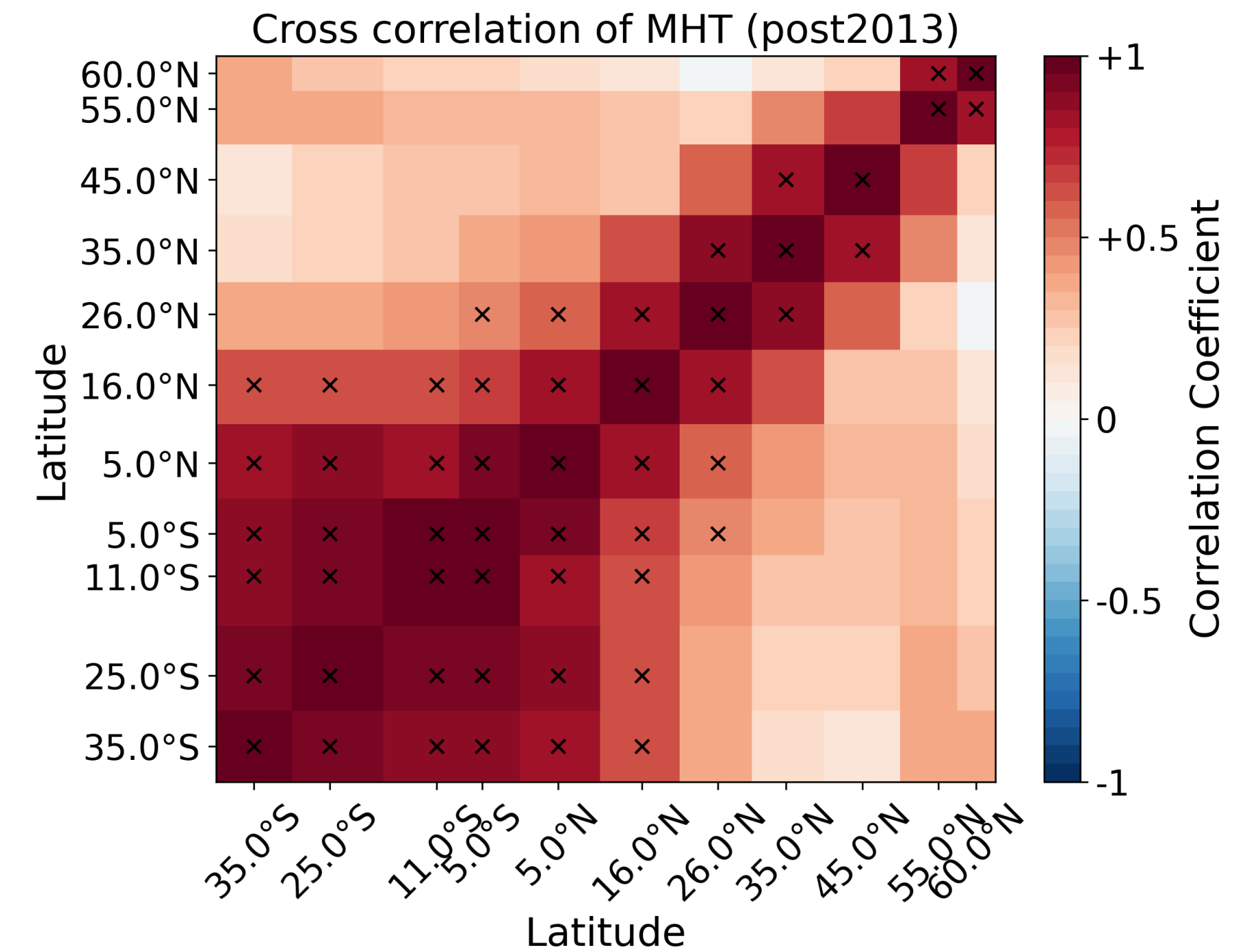
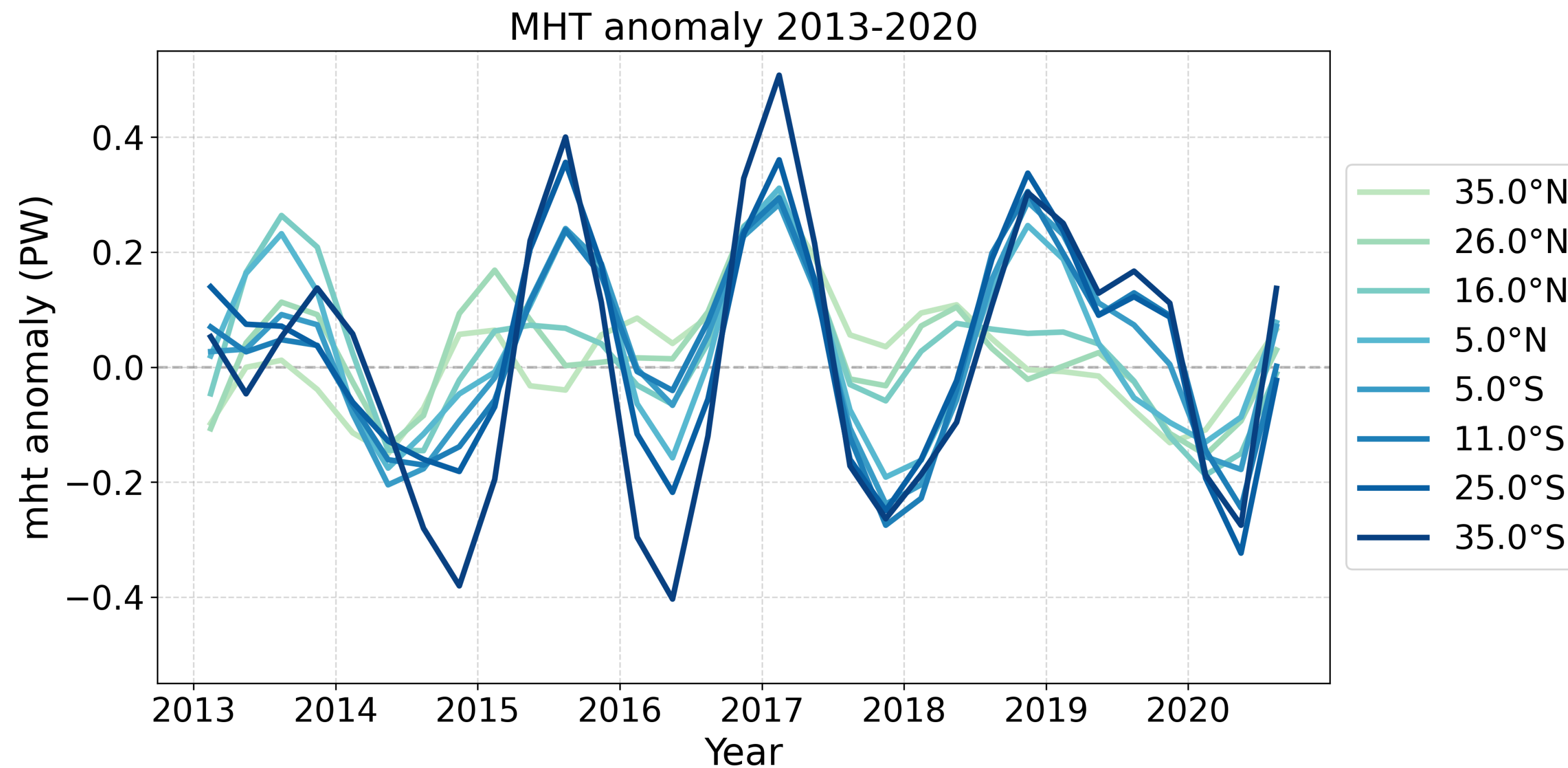
- Here we see signal goes from in phase to **anti-correlated**
- split into 2 covarying boxes:
 - 35 - 16°N (Subtropical block (**ST**))
 - 5°N - 35°S (South Atlantic block (**SA**))

MHT Coherence / correlation break 2009 - 2013



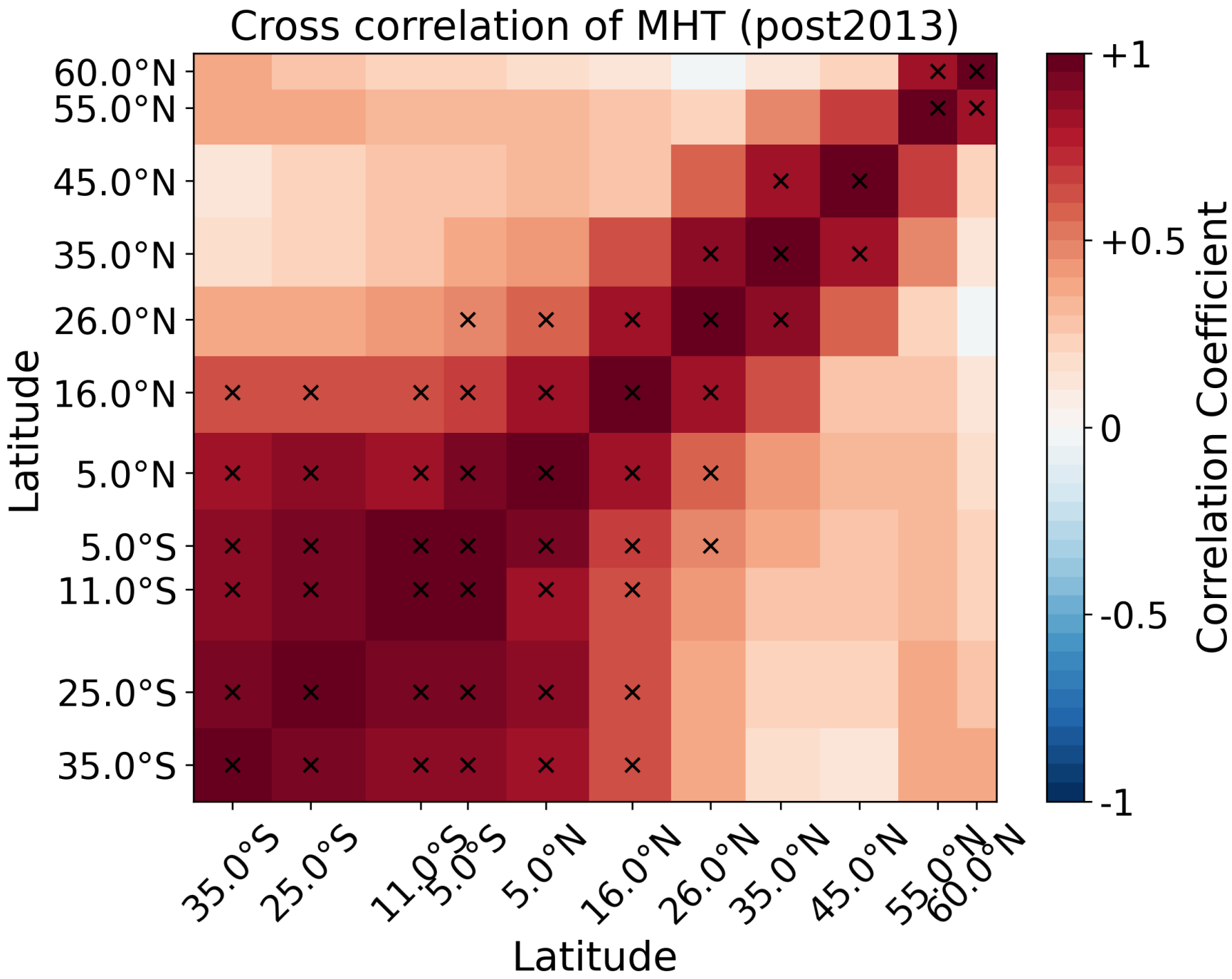
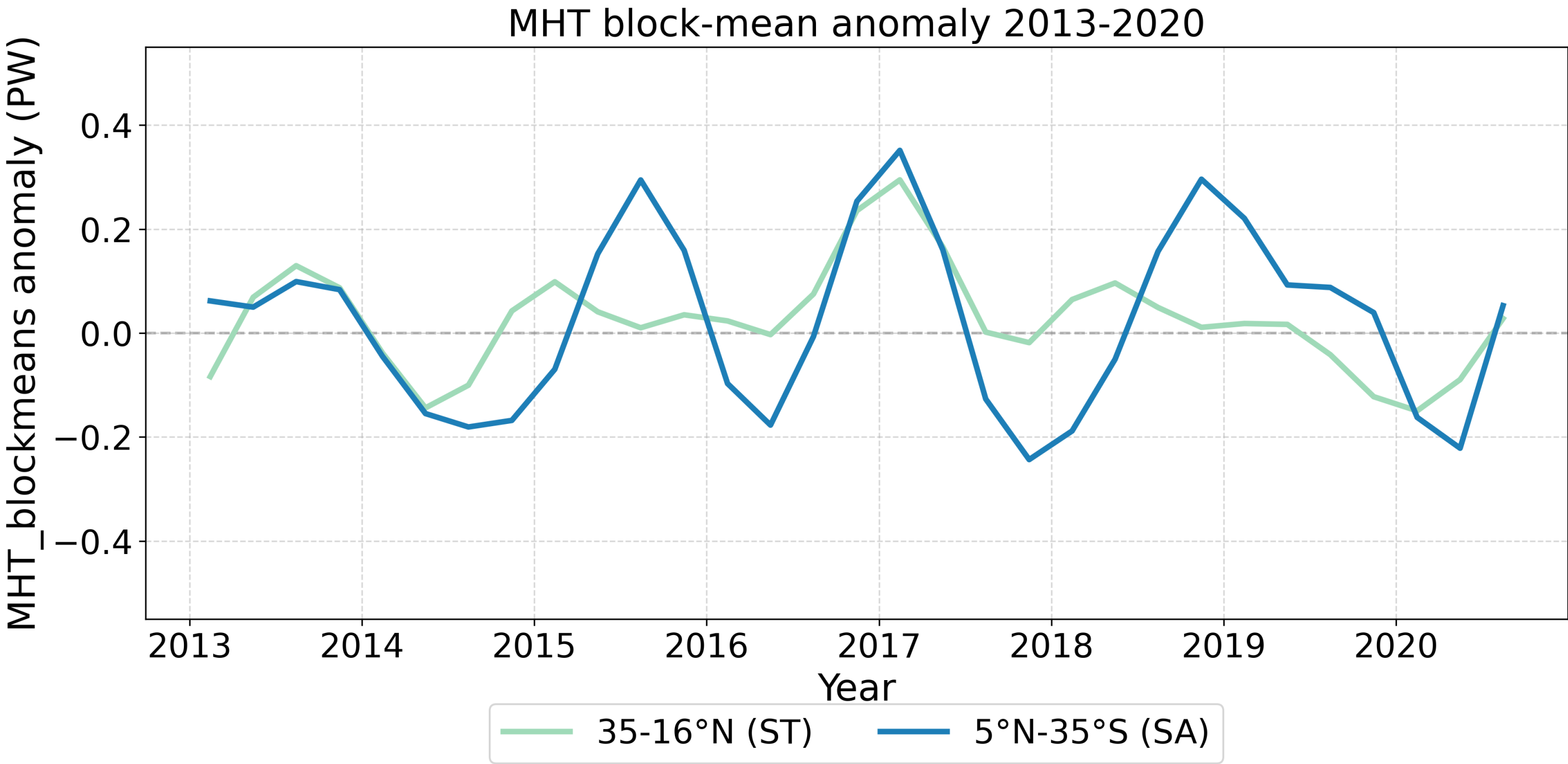
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MHT Coherence after 2013



- Again mostly **in phase signal** and **amplitude**, northern latitudes tend to have lower amplitudes

MHT Coherence after 2013



- Again mostly **in phase signal** and **amplitude**, northern latitudes tend to have lower amplitudes

Quantification of coherence break / split:

All in PW	Pre 2009	2009-2013	Post 2013
Amplitude ST	0.12	0.15	0.10
Amplitude SA	0.16	0.22	0.17
Spread ST	0.05	0.05	0.04
Spread SA	0.06	0.07	0.05
Spread across total block	0.10	0.14	0.08

Amplitude: block mean temporal std

- SA block has higher overall amplitudes and varies more strongly
- Both blocks show maximum amplitudes (ST: 0.15 PW and SA: 0.22 PW) in 2009-2013

Spread: latitudinal std in each period, mean over period

- Latitudes deviate more strongly in 2009-2013 across 35°N - 35°S
- dominated by spread in SA block

Quantification of coherence break / split:

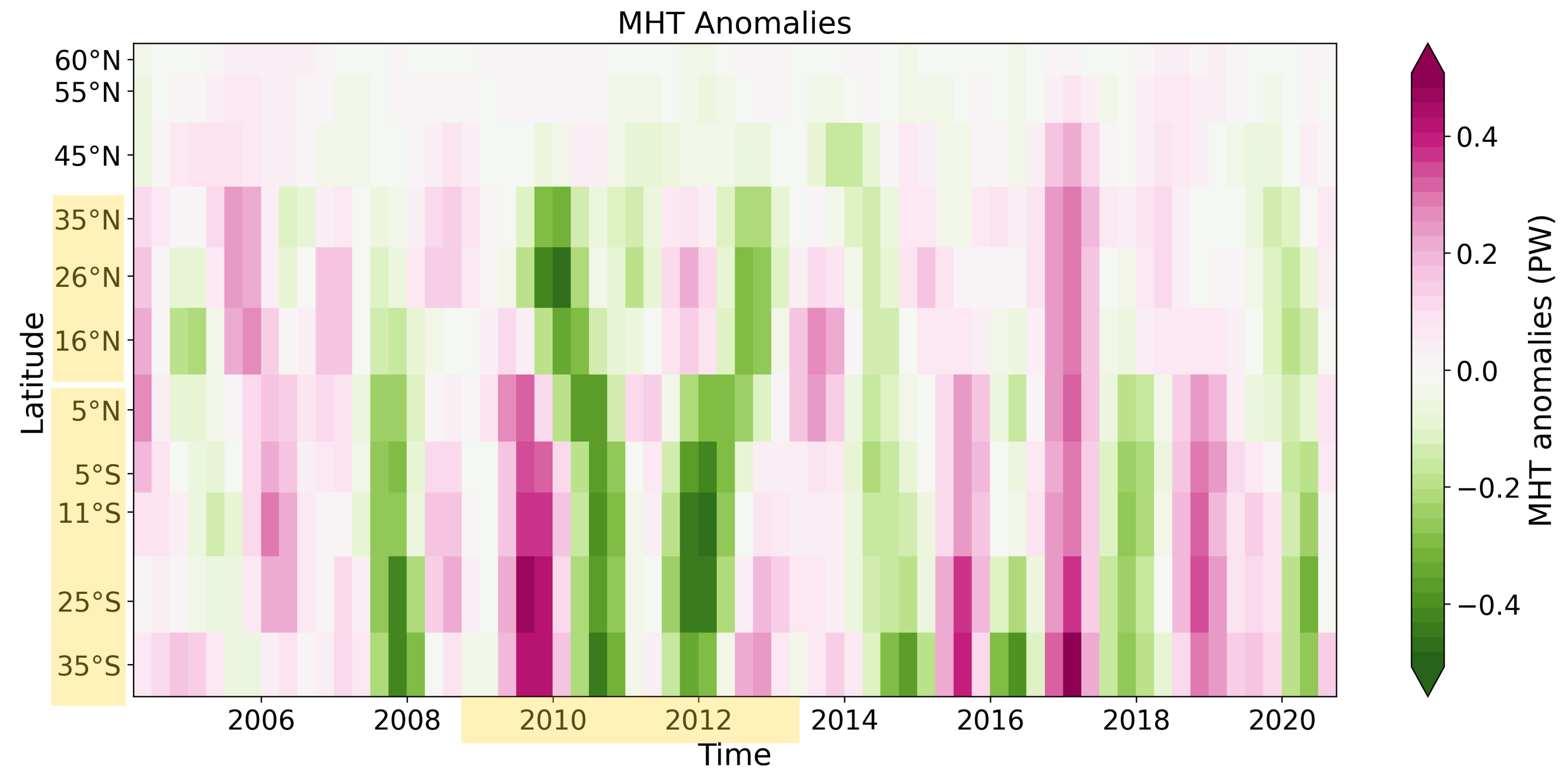
	Pre 2009	2009-2013	Post 2013
Corr between ST and SA block	0.54	-0.31	0.63
Corr within ST	0.80	0.90	0.85
Corr within SA	0.79	0.86	0.90

- 2009-2013 shows anti correlation between the ST (35 - 16°N) and SA (5°N - 35°S) block

Raw data again

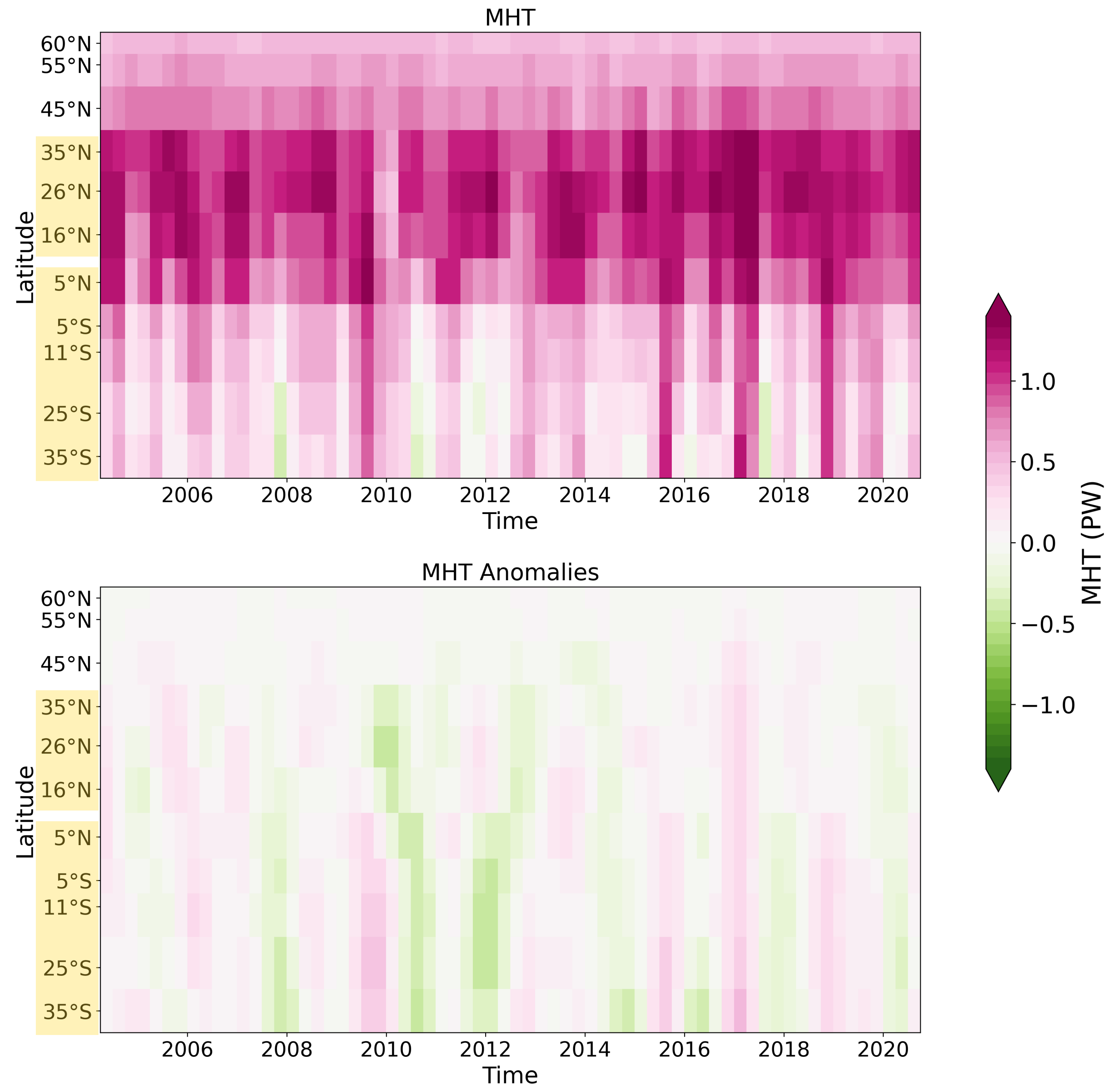
- Anomalies show split between 16 and 5°N
- 2010: strong negative anomalies arise first 35-16°N (ST), then later in south atlantic block (SA)
- 2012: negative anomalies in the SA block arise some quarters later in the ST block

→ pattern flips around

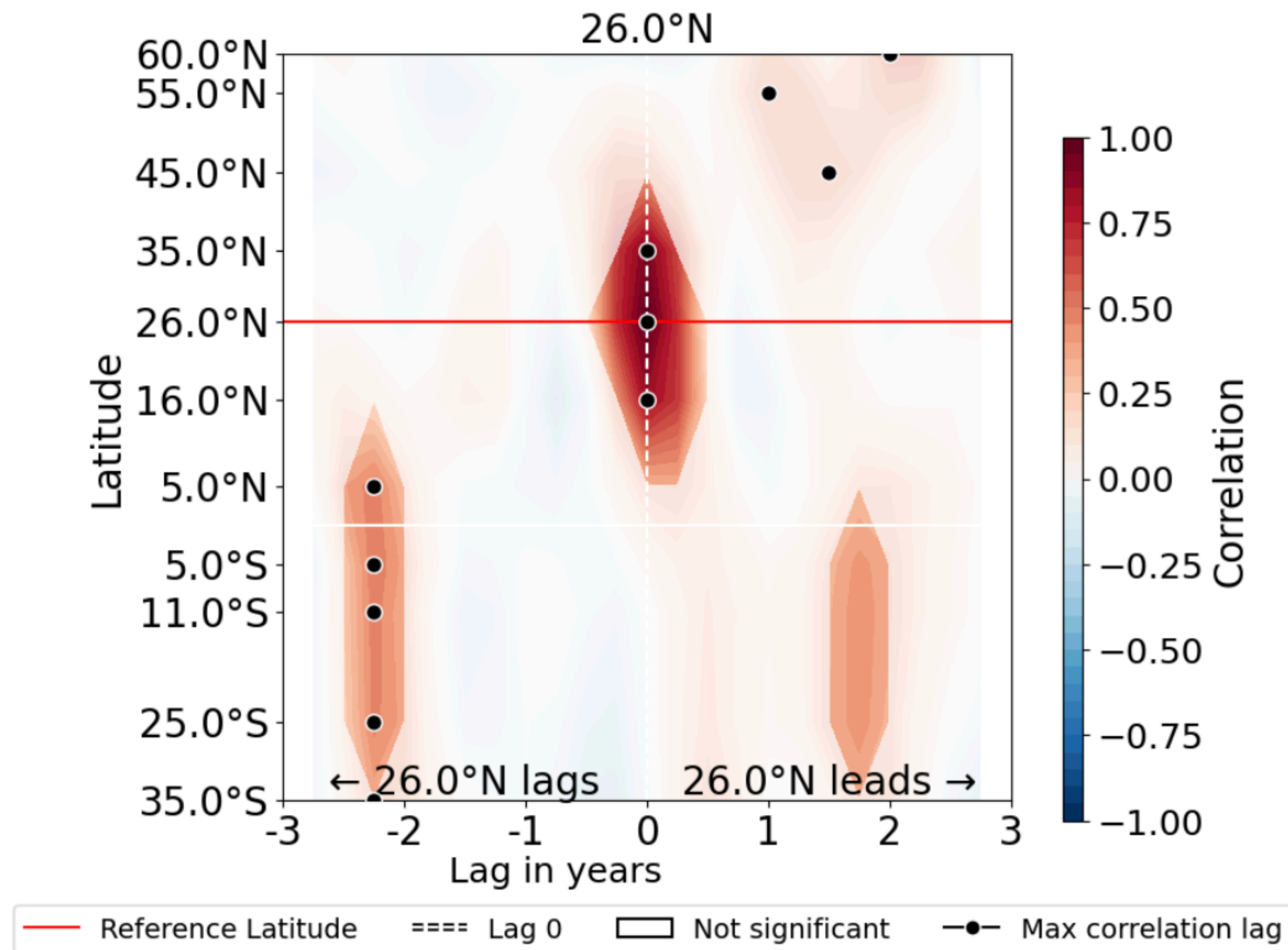


Raw data again

- Anomalies show split between 16 and 5°N
 - Looks like a propagation from 35-16°N to the more southern latitudes around 2010,
 - 2012: negative anomalies in the 5°N-35°S block arise some quarters later in the subtropical block
- pattern flips around



Lead / lag analysis of MHT anomalies



From lead / lag analysis only 26°N shows significant relationship:

- 35 - 16°N Band clearly visible
 - Lags 5°N - 25°S by ~2.5 yrs (maximum correlation lag)
- signal propagation south to north
- But also leads by ~2 yrs, less pronounced

Next steps / open questions:

Find propagation in the signal, do a max lag correlation plot

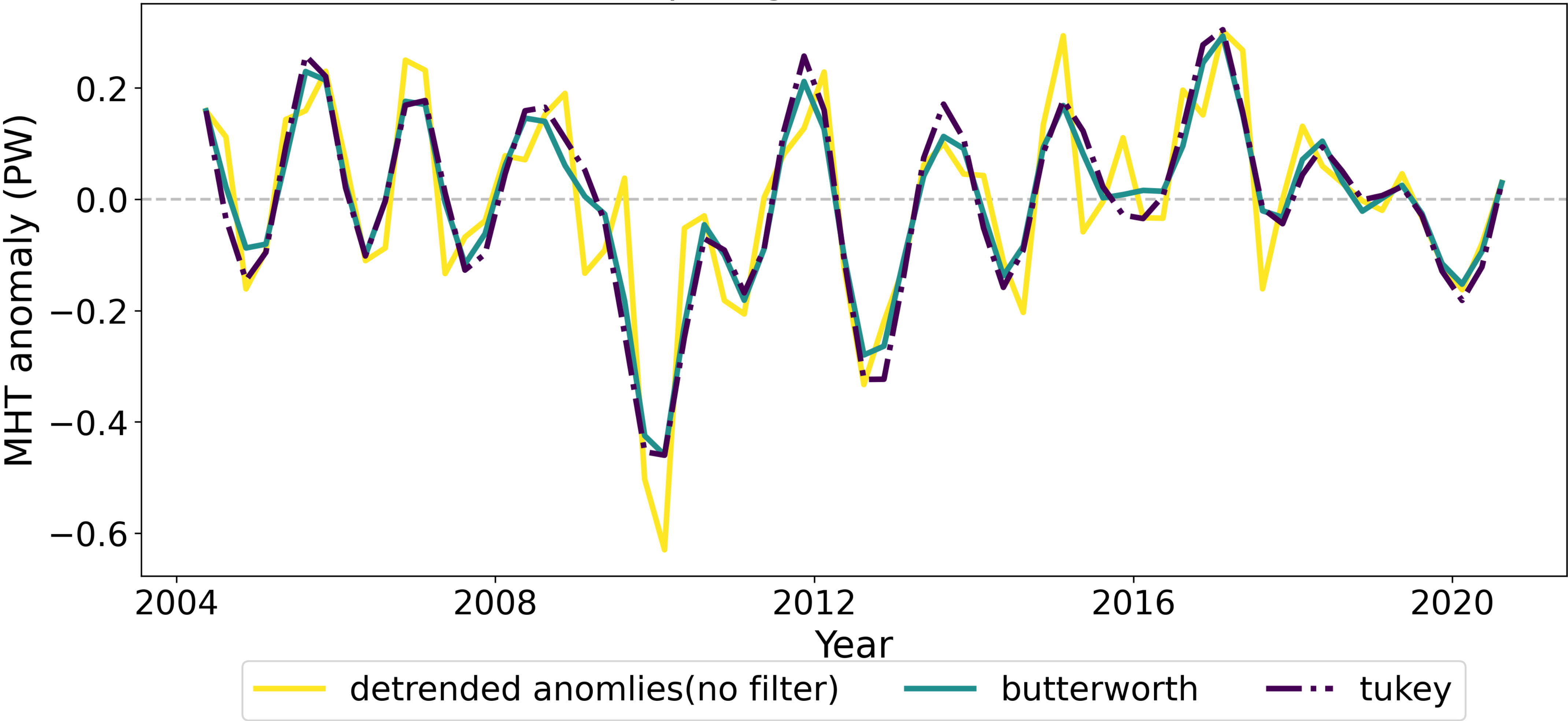
- Quantify the break down with in terms of how much heat and what timescales?

What caused the coherence break and split into two boxes?

- 2008/2009 record low NAO → AMOC & MHT drop
- Connect NAO - air sea heat flux F_S - MHT (HTC)

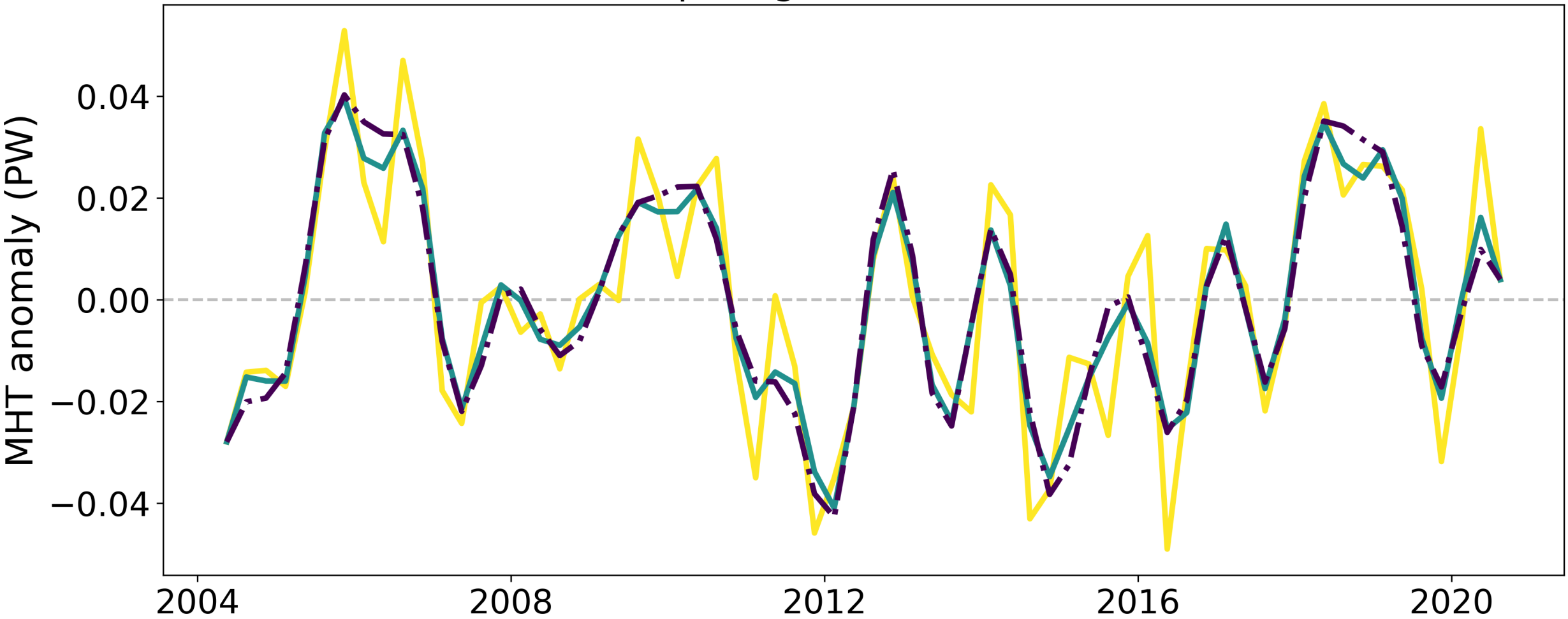
Processing, comparing filters

Comparing Filters at lat 26.0

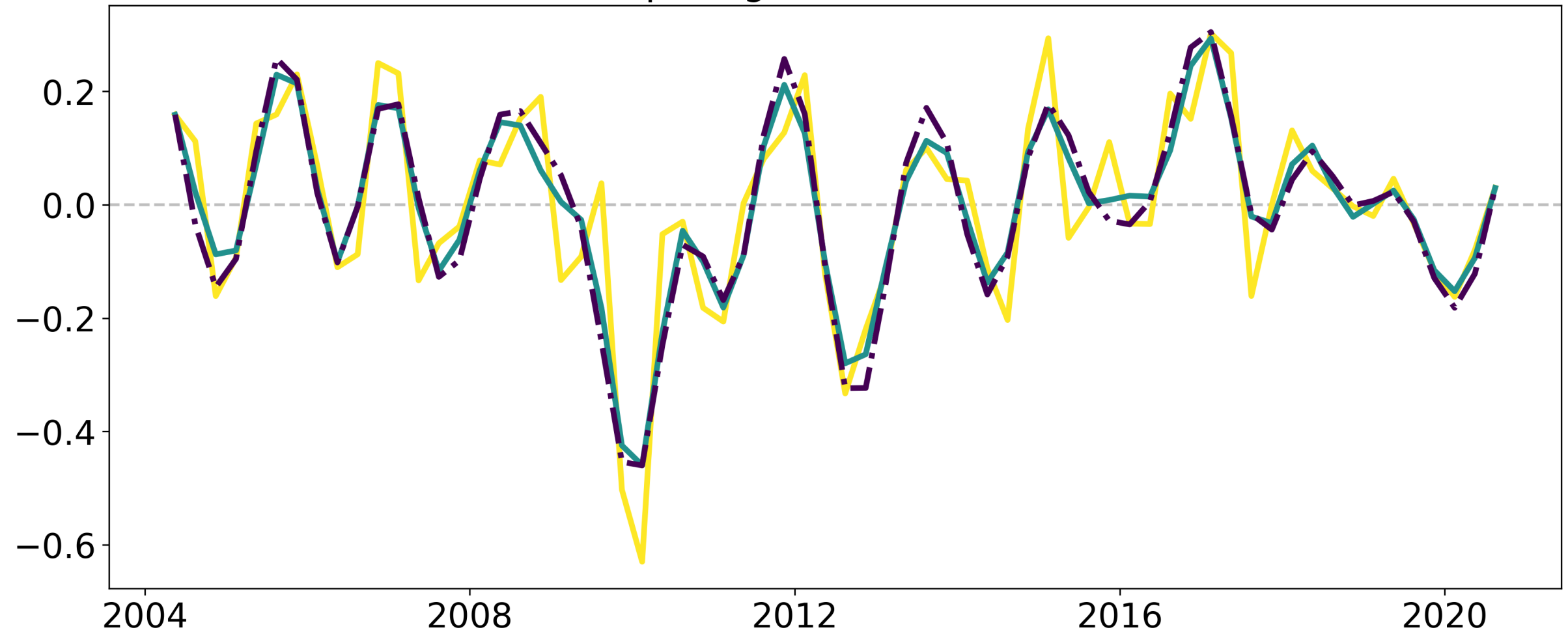


Processing, comparing filters

Comparing Filters at lat 60.0



Comparing Filters at lat 26.0



Comparing Filters at lat -25.0

